

Phaneendra Kumar Papinedi

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SUMMARY

Computer Science (IoT) graduate who has built end-to-end, real-time data systems: Python/OpenCV and YOLOv8 computer-vision pipelines, Raspberry Pi and ESP32 sensor platforms, and Pandas/NumPy analytics. Research Intern at IIIT Hyderabad and co-author of a DOI-indexed IoT paper. Looking to apply Python, SQL, data structures and AI/IoT engineering to large-scale client solutions at TCS.

TECHNICAL SKILLS

- **Languages:** Python, C, Java, SQL
- **Core CS (coursework):** Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks
- **Computer Vision and Data:** OpenCV, YOLOv8 (object detection), image processing, NumPy, Pandas, time-series data handling
- **IoT and Embedded:** Raspberry Pi, ESP32/ESP8266, ESP32-CAM, Arduino, sensor integration, data acquisition

EXPERIENCE

Research Intern, SPCRC, IIIT Hyderabad | Feb 2026 - Jul 2026

- Built an automated IoT pan evaporimeter for continuous, real-time monitoring of evaporation and environmental conditions.
- Developed a Python data-acquisition pipeline on Raspberry Pi that synchronises multiple environmental sensors for meteorological and evaporation measurements.
- Implemented periodic high-resolution image capture and Python/OpenCV pipelines to measure water-level changes in the evaporation pan from images.
- Cleaned, merged and analysed sensor and image-derived time-series data with Pandas and NumPy to identify evaporation patterns.

PROJECTS

Monkey Detection System | Python, YOLOv8, OpenCV

- Fine-tuned a pretrained YOLOv8n object detector for 50 epochs on a custom monkey dataset, reaching 0.82 mAP@0.5 and 0.54 mAP@0.5:0.95 on the validation split.
- Built a real-time inference script with OpenCV that runs the model on a webcam or video stream and overlays the detections with a live count.

Computer Vision Water Level Monitor | Python, OpenCV

- Automated lake water-level monitoring with an OpenCV pipeline, replacing manual observation with repeatable image analysis.
- Used region-of-interest segmentation and pixel-level measurement on reference markers to convert image data into water-level readings.

ESP32-CAM Attendance System | ESP32-CAM, backend server

- Built an embedded attendance system on ESP32-CAM that captures images on trigger and sends them to a backend server, removing manual attendance logging.
- Designed the upload and backend workflow that records and stores attendance data from the captured images.

PUBLICATION

Hand Gesture Recognition for Specially Impaired People Using IoT, IJRASET, Vol. 14, Issue IV, April 2026, pp. 3009-3016 (DOI: [10.22214/ijraset.2026.80001](https://doi.org/10.22214/ijraset.2026.80001)). Co-authors: Kancherla Balakrishna, P. Guru Manikanta, P. Sai, P. Kavitha.

- Co-authored a paper on a wearable, threshold-based (no machine learning) gesture recognition system using an ESP32, five flex sensors and an MPU6050.
- The paper reports 96% recognition accuracy and 78 ms average response time in prototype testing.

EDUCATION

- **B.Tech, Computer Science and Engineering (Internet of Things)**, RVR & JC College of Engineering, Guntur | 2022 - 2026 | CGPA 6.89/10
- **Intermediate (MPC)**, Bhashyam Junior College | 2020 - 2022 | 71.1%
- **Secondary Education**, Loyola Public School | 2019 - 2020 | 71.0%